Evaluating VLM Architectures for Low-Level Image Quality Diagnostics

DA624, NLP with Large Language Models, 2026

Course Project

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MOTIVATION

- **The Limitation:** Vision-Language Models (VLMs) demonstrate remarkable zero-shot capabilities in high-level semantics, but frequently fail at low-level perceptual tasks.

- **Traditional Inadequacy:** Standard Image Quality Assessment (IQA) relies on scalar metrics (PSNR, SSIM, LPIPS). They yield mathematical scores but lack interpretability.

PROBLEM STATEMENT

- **The Fusion Bottleneck:** The core challenge lies in the fusion mechanism—how visual features are aligned with textual embeddings without losing fidelity.

- **Investigation:** We aim to analyze how different multi-modal fusion strategies impact the network's ability to preserve and reason over high-frequency visual data.

FUSION STRATEGIES

1. **Early Fusion:** Concatenates raw image patch embeddings and text tokens early in the network through a joint transformer. Ex: ViLT

2. **Mid Fusion:** Uses a bridging mechanism (Q-Former) to extract visual tokens via cross-attention before generation. loss. Ex: BLIP-2

3. **Token-Level Fusion:** Employs an MLP projection layer to map ViT outputs directly into an LLM. Ex: LLaVA

DATASETS

1. **Q-Instruct:** Contains >200,000 QA pairs. Focused on image quality description, noise detection, and artifact reasoning for fine-tuning.

2. **DepictQA (M-BAPPS):** Targets descriptive IQA and comparative reasoning (e.g., evaluating a restored image strictly against its degraded input).

3. **Q-Bench:** Utilized strictly during inference to rigorously test the network's spatial grounding and exact artifact localization accuracy.

METHODOLOGY

1. **Baseline:** Set up open-source baselines representing targeted fusion strategies (ViLT, BLIP-2, LLaVA).

2. **PEFT Tuning:** Manage memory constraints using QLoRA. Update only cross-attention layers and LLM adapters on Q-Instruct.

3. **Inference & Error Analysis:** Models will be strictly evaluated using Q-Bench to gauge aggregate accuracy and spatial grounding. Conduct qualitative analysis to categorize failure modes.

TRAINING STRATEGY

- **Optimized for Constrained Environments:** To strictly manage GPU memory constraints while maximizing learning, we will employ Quantized Low-Rank Adaptation (QLoRA).

- **Primary vision encoders will be frozen:** Only the cross-attention/projection layers and select Large Language Model (LLM) adapter weights will be updated during training.

EARLY FUSION: VILT

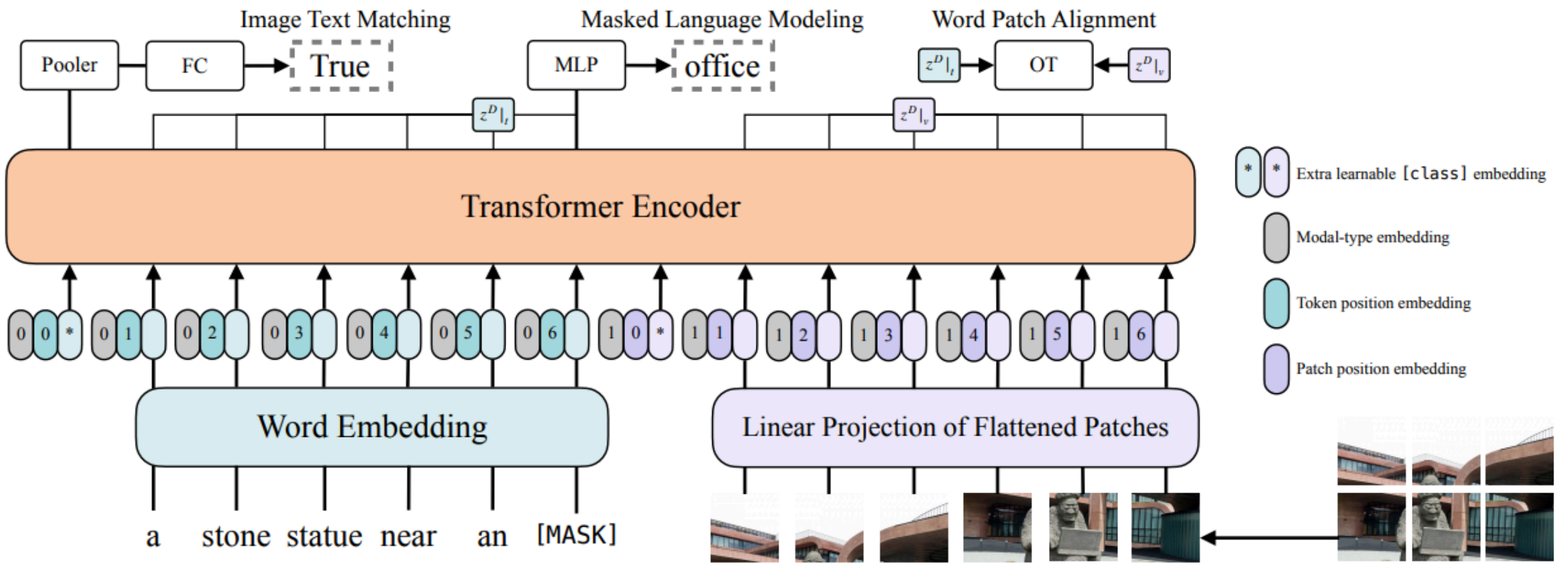


Figure 3. Model overview. Illustration inspired by Dosovitskiy et al. (2020).

- **One transformer for both:** image patches and text tokens go together into a single transformer from the start.
- **Light and simple:** no separate image encoder and no large language model. It answers by picking one word from a list of 3,129 answers.

EARLY-FUSION: SETUP & RESULTS

Model	All	Yes/No	What	How
Random Guess	37.8	50.0	-	-
Always “Yes”	-	57.1	-	-
Zero-shot ViLT	47.8	61.8	40.7	38.7
Fine-tuned (first run)	46.0	63.6	30.1	41.0
Fine-tuned (fixed labels)	47.8	62.9	35.8	42.0

Q-BENCH DEV SET, 1,495 QUESTIONS, ACCURACY IN %.

- 47.8% is about 10 points above random guessing (37.8%).
- Fixing the labels recovered What questions (30.1 to 35.8) compared with the first run.
- The fixed run matches zero-shot exactly: 715 of 1,495 correct.

EARLY FUSION ALONE DOES NOT SOLVE LOW-LEVEL QUALITY DIAGNOSIS WHEN THE OUTPUT IS A FIXED WORD LIST!

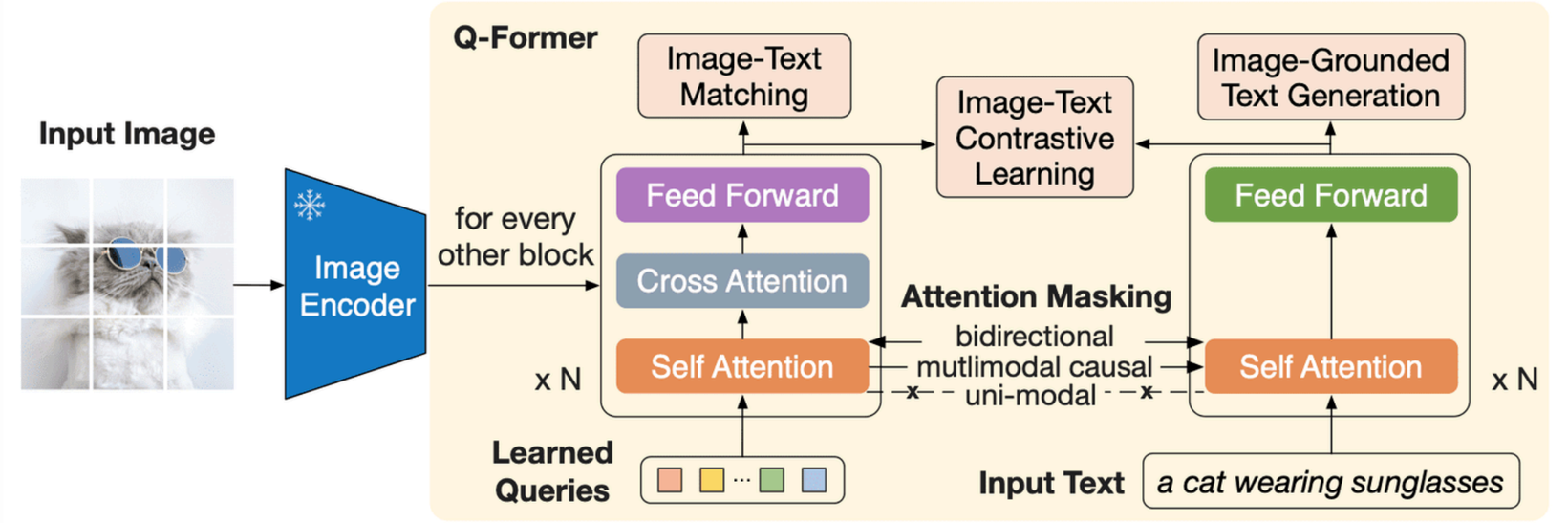
EARLY-FUSION: RESULTS

WHY VILT STRUGGLES / WHAT WE LEARNED

- **The answer format, not the fusion idea, is the main limit.**

1. **Fixed answer list (main limit)**
   - ViLT picks 1 word from 3,129 options.
   - Only 24% of Q-Instruct answers fit this list.
   - 6% of test questions (92) have no option it can score.
2. **Weak on detail and reasoning**
   - Yes/No: 62.9%, only slightly above “always Yes” (57.1%).
   - The model leans toward “no”: it says “yes” on about 40% of Yes/No questions, but 57% of answers are “yes”.
   - Local distortion is the hardest group (43.1%, against 49.8% for global distortion). This is a hint, not a proof.
3. **Task mismatch**
   - ViLT is built for short, meaning-based answers (VQA).
   - Image quality needs fine detail and descriptive answers.

MID-FUSION: BLIP-2



BLIP-2 Q-Former Pre-training

- 257 image tokens are compressed to 32 learned queries.
- Q-Former is trained only on image-caption objectives.
- The bottleneck discards the low-level detail, and Q-Former must be fine-tuned to recover it

MID-FUSION: SETUP & RESULTS

Model	All	Yes/No	What	How
Zero-Shot	42.4	54.6	36.5	34.3
Fine-tuned	55.3	68.5	45.5	49.7

Training Setup

- Dataset: Q-Instruct
- No. of Samples: 14k
- LoRA on OPT-2.7B
- 45M Trainable params

Testing Setup

- Benchmark: Q-Bench
- No. of Questions: 1.5k

MID-FUSION: RESULTS

Trained	Q-Bench
LoRA only (Q-Former Frozen)	51.9
+ cross-attention	55.3
+ full Q-former	54.8

- Grey-Image Test: Zero-shot loses only 2.5 points, whereas finetuned loses 12.8 points
- The image matters 5x more

- Botteleneck Test: Distortion type can be read from the features 76% of the time before the Q-Former, but only 56% after.
- Training the Q-Former increases the performance by 3%.

MID-FUSION: CAN IT EXPLAIN?

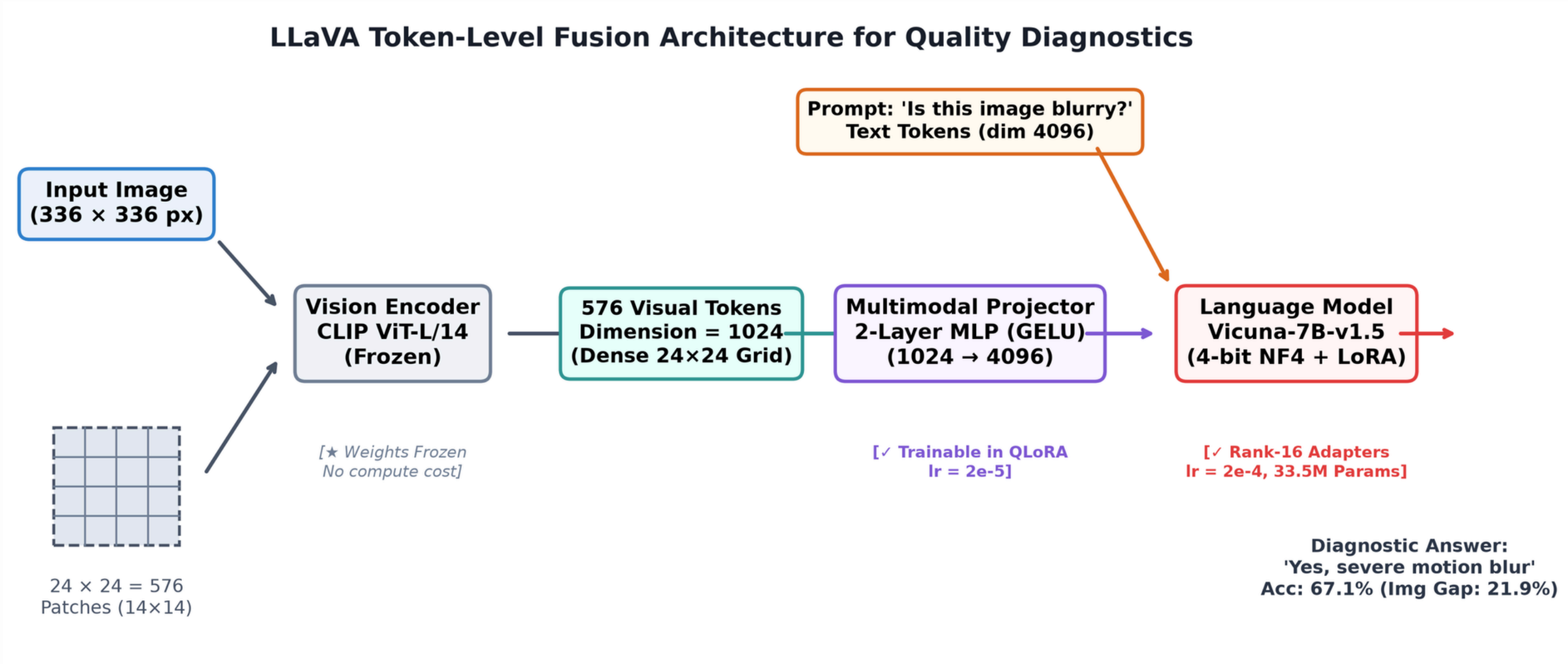


Added Ringing (L3) artifacts

- Zero-Shot: It describes the content, but not the quality. It calls a “A digital painting of hay bales in the field”.
- Fine-tuned: It gets the severity right, but not the actual cause. The noise is described as “Serious focusing issues”.

Takeaway: Answering ≠ Explaining

LATE-FUSION (LLAVA)



- Maps every ViT token directly into the language model via a 2-layer MLP projection.
- Keeps all 576 visual tokens, avoiding the bottleneck that discards low-level details.

LATE-FUSION: SETUP & RESULTS

Model	All	Yes/No	What	How
Zero-Shot	58.7	66.4	58.2	50.5
Fine-tuned	67.1	76.2	66.4	57.6

Training Setup

- Dataset: Q-Instruct
- No. of Samples: 14k
- LoRA on Vicuna-7B (4-bit NF4)
- 33.5M Trainable params

Testing Setup

- Benchmark: Q-Bench
- No. of Questions: 1.5k

LATE-FUSION: RESULTS

Trained	Q-Bench
Projector only (LLM Frozen)	61.2
LoRA only (Projector Frozen)	63.8
Projector + LoRA (Default)	67.1

- Grey-Image Test: Zero-shot loses 15.6 points (43.1%), whereas finetuned loses 21.9 points (45.2%).
- The image matters most in token-level fusion

- Preservation Test: 576 tokens yield +15.8% on Global Distortion (65.2%) and +10.2% on Local Distortion (64.8%).
- Adapting both projector and LLM increases performance by 8.4 points.

COMPARISON MATRIX: ALL 3 FUSION TYPES

EVALUATED ON Q-BENCH DEV (N = 1,495 QUESTIONS; RANDOM BASELINE = 37.8%)

- **EARLY FUSION (VILT-B/32):**
  ARCHITECTURE: JOINT VIT TRANSFORMER | TOKENS: 32×32 PATCHES (COARSE) | PARAMS: 111M
  SCORES: ZERO-SHOT: 47.8% → FINE-TUNED: 47.8% (NET GAIN: +0.0%, IMAGE GAP: N/A)
  KEY LIMIT: FIXED 3,129-WORD CLASSIFICATION HEAD COMPLETELY PREVENTS QUALITY ADAPTATION.

- **MID FUSION (BLIP-2 OPT-2.7B):**
  ARCHITECTURE: Q-FORMER CROSS-ATTENTION | TOKENS: 32 LEARNED QUERIES (14.7× DROP) | PARAMS: 3.77B
  SCORES: ZERO-SHOT: 42.4% → FINE-TUNED: 55.3% (NET GAIN: +12.9%, IMAGE GAP: 12.8%)
  FINDING: RECOVERS DISTORTION AWARENESS, BUT 32 QUERY TOKENS BOTTLENECK FINE LOCAL TEXTURES.

- **TOKEN-LEVEL FUSION (LLAVA-V1.5-7B)**:
  ARCHITECTURE: 2-LAYER MLP PROJECTOR | TOKENS: 576 VISUAL TOKENS (DENSE 24×24 GRID) | PARAMS: 7.06B
  SCORES: ZERO-SHOT: 58.7% → FINE-TUNED: 67.1% (NET GAIN: +8.4%, IMAGE GAP: 21.9%)
  TAKEAWAY: SUPERIOR ACCURACY, DEEPEST VISUAL RELIANCE (21.9%), PRESERVES FINE LOCALIZED ARTIFACTS.

KEY TAKEAWAYS:

1. Visual Token Conservation is Decisive

• BLIP-2's 32-query bottleneck discards high-frequency detail (Answering != Explaining).

• LLaVA keeps all 576 tokens, preserving subtle local ringing, noise, and texture loss.

2. Output Space Flexibility Dictates Utility

• ViLT's fixed word list yields zero gain (+0.0%), proving classification heads cannot do IQA.

• Autoregressive decoder LLMs are essential for causal, explainable diagnostic reports.

3. Visual Grounding Spectrum

• Image Gap scales monotonically:

• Token-level projection compels the LLM to genuinely inspect raw pixel features.

CONCLUSION

• Token-Level Fusion (LLaVA) achieves superior diagnostic performance:

• Dense Visual Pathway Preserves Fine Artifacts

• Maximum Perceptual Grounding:
Expands blind control Image Gap to 21.9% (vs BLIP-2's 12.8%), demonstrating deepest reliance on image evidence over language priors.

• Eliminates Semantic Hallucination:
Q-Instruct fine-tuning grounds the LLM, transforming aesthetic rationalizations (e.g. 'digital painting') into factual diagnostic reports.

LIMITATIONS

- COMPUTATIONAL FOOTPRINT
- TASK & BENCHMARK SCOPE
- PRE-TRAINING SEMANTIC BIAS

FUTURE WORK

- ADAPTIVE MULTI-SCALE TOKEN PRUNING
- IN-THE-LOOP DIAGNOSTIC PERCEPTUAL CRITIC

THANK YOU!